# Bellamkonda Harshavardhan

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## SKILLS

- **Languages**: Python, Java, SQL, OOP,REST APIs
- **Generative AI:** LLMs, Prompt Engineering, RAG, LangChain, LangGraph, Middleware, In-Memory Memory, OpenAI APIs, AI Chatbots, Embeddings, Vector Search
- **Agentic AI:** AI Agents, LLM Orchestration, Tool Calling, API Integration, Agent Workflows, Conversational Memory, Multi-Step Workflows
- **AI/ML:** Machine Learning, Deep Learning, Neural Networks, Transfer Learning, TensorFlow, Keras, ResNet50, Model Evaluation, Feature Engineering, EDA,
- **Data & Analytics:** MySQL, MongoDB, Power BI, Tableau, Excel, PySpark, Data Cleaning, Data Validation, Data Quality, Reporting
- **Cloud & Development:** Microsoft Azure, Azure App Service, Azure SQL, Git, GitHub, Terraform, Flask, Agile/SDLC

## Professional Summary

Computer Science Engineering graduate (2026) with hands-on experience in Python, Machine Learning, Deep Learning, Generative AI, NLP concepts, RAG, prompt engineering, and AI application development. Built LLM-powered applications using LangChain, OpenAI APIs, CHROMA, and FastAPI, including a document question-answering RAG system with vector retrieval. Experienced with TensorFlow/Keras, model evaluation, data preprocessing, REST APIs, SQL, MongoDB, Azure, and web application integration. Strong foundation in building, evaluating, and deploying practical AI solutions, with an interest in Agentic AI, LLM orchestration, tool integration, and scalable AI workflows.

## INTERNSHIP

### Data Analyst Intern | Bluestock Fintech — May 2026 – Jul 2026

Worked on financial and business datasets using Excel, SQL, and Python to identify trends and generate useful insights. Gained hands-on experience in data cleaning, preprocessing, validation, and exploratory data analysis (EDA) to improve data quality and reporting accuracy. Created interactive dashboards and visual reports using Power BI and Excel and prepared performance reports to support data-driven decision-making. Collaborated with the project manager and team members to complete analytical tasks within project timelines.

## PROJECTS

### NexusIQ – Enterprise Multi-Agent AI Copilot | LangGraph | Link

Technologies: Python, LangGraph, LangChain, OpenAI API, ChromaDB, RAG, MCP, Text-to-SQL, SQLite, Streamlit, Docker

- Built a multi-agent AI copilot using LangGraph, with a supervisor that routes each question to a RAG agent, a SQL agent or a tool-calling agent, and runs agents in parallel for combined questions.
- Implemented hybrid RAG on ChromaDB (vector search and BM25 merged with Reciprocal Rank Fusion, heading-aware chunking), returning answers with source citations and a self-critic step to reduce unsupported responses.
- Developed a secure text-to-SQL agent with sqlglot query validation, a read-only database connection, role-based column access control and automatic self-correction of failed queries.
- Exposed the system as an MCP server with 5 tools and built Streamlit and Chainlit interfaces with live agent traces and LangSmith tracing; added prompt-injection and PII guardrails.

### Dibe Retinal – Explainable AI Platform for Diabetic Retinopathy Detection | Link

Technologies: Python, TensorFlow, Keras, ResNet50, Grad-CAM, OpenCV, Flask, HTML, CSS, JavaScript, MySQL

- Developed an end-to-end AI platform for automated diabetic retinopathy detection from retinal fundus images using deep learning.
- Implemented a ResNet50-based transfer learning model to classify retinal images across five severity levels with explainable AI support through Grad-CAM heatmaps.
- Built a secure web application featuring user authentication, image upload, AI inference, prediction history tracking, and PDF generation
- Designed automated reports containing risk assessment, confidence scores, precautions, and diet recommendations.

### Predictive Analysis with Machine Learning Models | Link

Technologies: R, Scikit-learn, Machine Learning, Data Preprocessing, Model Evaluation

- Implemented KNN, Linear Regression, Decision Trees, Random Forest, and SVM models for predictive analysis.
- Focused on data preprocessing, predictive modeling, and performance evaluation.

## CERTIFICATES

- Microsoft Certified: Azure Fundamentals(AZ – 900) | Microsoft — **July-Aug 2024**
- SupervisedMachine Learning: Regression and Classification | Coursera — **Oct-Nov 2025**
- SAP Certified Associate – Data Analyst – SAP Analytics Cloud | Sap — **Feb 2026**

## EDUCATION

**Lovely Professional University Punjab** — **2022 – 2026**
*Computer Science and Engineering -CGPA : 7.2* — *Jalandhar, Punjab*

**SriChaitanyaJunior College** — **2020– 2022**
*12th with Science — Percentage: 92* — *Khammam, Telangana*

**New era School** — **2019 – 2020**
*10th with Science — CGPA: 9.8* — *Khammam, Telangana*